

Shlok Joseph Paul

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Research Overview

A chemist and chemical engineer specializing in colloidal quantum dot (cQD) photodetectors for near- to mid-wave infrared (NIR–MWIR) sensing. During a DARPA- and ONR-funded PhD at NYU, I led the development of an end-to-end cQD infrared detection thrust in our lab. Starting from optimizing cQD synthesis to fabricating photodetectors and setting up automated testing systems.

My three main scientific contributions were designing a RoHS-compliant synthesis of Ag₂Se quantum dots coupled with a solution-phase ligand exchange to obtain infrared responsive inks. Engineering a three-stage automated dip-coater that leverages a biphasic coating approach to dramatically boost fabrication throughput while minimizing material consumption, and finally developing an infrared-transparent silver nanowire – polymer composite for use as an electrical contact for top illuminated or non-planar photodetectors.

Now at Los Alamos I am focusing on understanding how to harness and then apply hot electrons in colloidal semiconductors.

Education

New York University – Tandon School of Engineering [07/2020 – 12/2025] Position: PhD Student (PI: Dr. Ayaskanta Sahu)

Doctor of Philosophy in Chemical and Bio-Molecular Engineering || Cumulative GPA: 3.94/4.0

Thesis Title: A Path to Scalable and Sustainable Production of Colloidal Quantum Dot Infrared Detectors.

New York University – Tandon School of Engineering [08/2015 – 05/2020] Position: UG Student

Bachelor of Science in Chemical and Bio-Molecular Engineering || Major GPA: 3.93/4.0

New York University – College of Arts and Sciences [08/2015 – 05/2020] Position: UG Student

Bachelor of Science in Chemistry || Cum Laude || Minor in Mathematics || Cumulative GPA: 3.53/4.0

Work Experience

Los Alamos National Lab: Postdoctoral Research Fellow – C-PCS Victor Klimov Group || Los Alamos, NM: March 02 - Present

- **Hot Electron Transport:** Understanding the role of Mn ions in the regeneration of hot electrons for photoelectrochemical, energy storage and ultrafast photodetection applications.
- **Conductive and Emissive Inks:** Applying solution phase ligand exchange protocols to highly emissive quantum dots to maintain their emission while improving quantum dot coupling.

Hybrid Nanomaterials Lab: PhD Research Assistant – Ayaskanta Sahu Lab Group || New York, NY: May 2020 – Feb 2026

- **Publications:** First author on two peer-reviewed papers and one manuscript under review, plus co-author on nine additional peer-reviewed articles spanning detectors, thermoelectrics and catalysis use cases of colloidal quantum dots.
- **Transparent IR Electrodes:** Designed a flexible silver-nanowire electrode that maximizes infrared photon transmission and electrical conductivity, enabling low-cost, scalable top-illumination for infrared ROICs.
- **SPLE for Ag₂Se QDs:** Developed a RoHS-compliant solution-phase ligand-exchange protocol for Ag₂Se quantum dots, yielding highly passivated films with enhanced reproducibility and record detectivity in NIR photodiodes.
- **Automated Dipcoating:** Built a custom automated dipcoater, slashing photodetector prototype iteration and testing time by ~87% to meet DARPA's rapid-prototyping requirements.
- **Biphasic Dipcoating Innovation:** Pioneered a biphasic dipcoating strategy that significantly reduced quantum-dot consumption per device, cutting material usage and costs.
- **High-Throughput Characterization:** Engineered a fully automated UV–LWIR LED/monochromator system to measure 3 dB bandwidth, rise/fall time, responsivity, and LDR— enabling processing of >50 devices/week.
- **Single-Pixel NIR Imaging:** Developed a single-step scanner using lab-fabricated single-pixel detectors to capture 2D NIR images.
- **Mentorship:** Supervised 15+ undergraduates and 15+ high-school students on quantum-dot detector projects, fostering hands-on research experience.
- **Conference Presentations:** Delivered invited talks and posters at major conferences in the US (3,000+ attendees) and Europe (100+ attendees), earning multiple presentation awards.

Entegris: Manufacturing Intern – Specialty Chemicals and Engineered Materials Group || Danbury, CT: May 2019 – Aug 2019

- **LabVIEW Automation:** Implemented a LabVIEW-driven pressure-regulator testing sequence, increasing regulator production throughput by 23% and automated the operator workflow using DAQExpress reducing cycle wait times by 83%.

- **Gas-Blend Cost Analysis:** Evaluated a 5% H₂/95% N₂ mixture for leak detection operations, cutting gas consumption costs by a factor of 12 and created pressure-correlation models in minitab to benchmark leak rates of He versus H₂N₂.
- **Process Change Leadership:** Directed Change FMEA and Validation & Qualification meetings for two key manufacturing process upgrades.

NYU Molecular Design Institute: Undergraduate *Research Assistant – Sacanna Lab Group* || New York, NY: Jan 2018 – Oct 2018

- Incorporated hollow TPM microcapsules into an adhesive thin-film formulation and demonstrated a 6 °C temperature reduction in a confined-system heat-trapping experiment.
- Awarded the Société de Chimie Industrielle Fellowship for innovative work on air-trapping colloids in heat-trapping paint applications.

Glenmark Pharmaceuticals: *Lab Summer Intern – Antibody Discovery Lab I* || Mahape, India: May 2017 – Jul 2017

- Performed Electroporation experiments to improve the permeability of target cells to ligated plasmids.

NIH National Cancer Institute: *Lab Assistant – Laboratory of Immune Cell Biology* || Bethesda, MD: Jul 2016 – Aug 2016

- Performed various introductory lab techniques such as Electrophoresis and Western Blots.

Selected Publications (Total 14 Peer Reviewed | Citations: 146 | H-Index: 6)

1. **Paul, S.J.**, Li, Letian., Vataj, A., & Sahu, A. Silver Selenide Colloidal Quantum Dot Inks for Heavy-Metal Free Infrared Photodetection. *ACS Applied Materials and Interfaces*. <https://pubs.acs.org/doi/10.1021/acsami.5c12011>
2. Park, S.Y, **Paul, S.J.**, Kang, M.S, Sahu, A. Direct Photopatterning of Heavy-Metal-Free Colloidal Quantum Dots for Multispectral Infrared Photodetectors. *Advanced Functional Materials*. <https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.202512679>
3. **Paul, S.J.**, Li, R., Kim, A., Li, Z., Sahu, A. Biphasic Meniscus Coating of Colloidal Quantum Dots for Scalable and Automated Infrared Photodetector Fabrication. **[Submitted: Small]**
4. **Paul, S. J.**, Molnas, H., Farrell, S., Parashar, N., Riedo, E., & Sahu, A. (2024). Plenty of Room at the Top: Exploiting Nanowire – Polymer Synergies in Transparent Electrodes for Infrared Imagers. *Journal of Materials Chemistry C*. <https://doi.org/10.1039/D5TC00581G>

Patents

1. “Product, System and Method with Silver Nanostructures Thin Film for Infrared Photodetector” US/PTO No: 63/519,406

Conference Talks and Posters

1. **(Best Poster Award) Poster and Talk:** “Solution Processed RoHS-Compliant Ag₂Se Quantum Dot Inks and Ag-Nanowire Matrices for NIR-SWIR Imaging with Potential in MWIR Applications” *30 Years of Quantum Dots at Los Alamos 2025*
2. **Poster:** “Shining a Light on Heavy Metal Free Quantum Dot Inks for Near to Mid-Wave Infrared Detection” *ASRC 2024*
3. **Poster:** “Heavy Metal Free Colloidal Quantum Dot Inks for Near Infrared Detection” *MRS Fall 2024*
4. **Talk:** “A Path to Heavy Metal-Free Infrared Photodetection Using Silver Selenide Colloidal Quantum Dots” *MRS Fall 2022*
5. **Poster:** “Heavy Metal-Free IR Photodetection using Ag₂Se Colloidal Quantum Dots” *GRC 2022*
6. **(#3 Place Poster Award) Poster:** “Synthesizing Silver Selenide Thin Films Via the Cation Exchange Process for Thermoelectric Applications” *AiChE 2019*
7. **(Quarter-finalist) Talk:** “3D-printing to augment research in lab” Stern 300K Entrepreneurship Challenge – New Product Ventures 2018
8. **Talk:** “A novel index for the early detection of thalassemia minor” (<https://www.ted.com/tedx/events/13227>) TEDxJNS 2015

Lab Skills: NMR, FTIR, UV-Vis, PL, XRD, SEM/EDX, TEM, AFM, Nanofabrication, XPS, Lock-In Amplifiers, Noise Spectral Density Measurements, 3dB Bandwidth and Response Time Characterization, Quantum Dot Synthesis.

Computer Skills: Origin, Python, MATLAB, Polymath, SolidWorks, LabVIEW and Minitab.

Relevant Classes: Project Management, Theory of Semiconductors, Nanomaterials Characterization, Numerical Methods

Awards

Société de Chimie Industrielle Research Fellowship

- Awarded for work done in manipulating TPM colloids to act as heat insulators in a polymeric matrix thin film.

NYU Awards

Deans Research Fund Awards (2018,19,20), Founders Day Award, Dean’s List (2018,19,20) and Heaviest Bench Press (2020)